

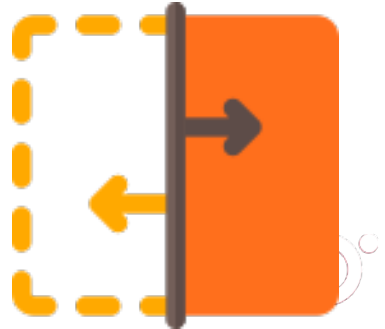
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The Snowflake Data Platform



Prerequisites

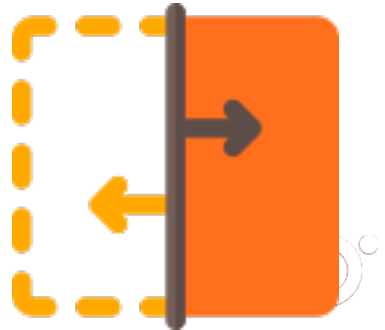
- Comfortable writing SQL queries to analyze data
- Some familiarity with working on the cloud



Set up for demos

- Please sign up for a free Snowflake account
<https://signup.snowflake.com/>
- Google Drive link for resources

https://drive.google.com/drive/u/3/folders/1Quk25l4NYp-4tk_GjK6qHKDm8NG4i7fh



General Poll

How comfortable are you with SQL?

- Never written queries in SQL before
- Somewhat comfortable writing SQL queries
- Very comfortable writing SQL queries





General Poll

How much experience do you have working on cloud platforms?

- No experience at all
- Somewhat comfortable working on the cloud
- Very comfortable working on the cloud



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The Snowflake Data Platform



Snowflake

Snowflake is a cloud-native, software-as-a-service advanced data platform for data storage, processing and analytics



Snowflake

Snowflake is a **cloud-native, software-as-a-service** advanced data platform for data storage, processing and analytics



Data Platform as a Cloud Service

Not hardware to
setup, install,
configure, or
manage

No software to
install, configure,
or manage

All administration,
maintenance,
upgrades
managed by
Snowflake



Data Platform as a Cloud Service

Snowflake runs entirely on the public cloud infrastructure

Uses VMs for compute and cloud storage services for persistent storage

Users **cannot** set up software and use Snowflake on private, on-premises clusters



Snowflake

- Does not use any existing database technology
 - e.g. MySQL, PostgreSQL, SQL Server
- Is not built on top of any existing big data framework
 - e.g. Hadoop, Spark
- Proprietary big data SQL engine
- Innovative architecture designed for the cloud



Where Does Snowflake Run?



Supported Cloud Platforms

- On each platform Snowflake provides one or more regions where the account is provisioned
- Can choose to host all Snowflake accounts on the platform where you run other cloud services



Snowflake Architecture

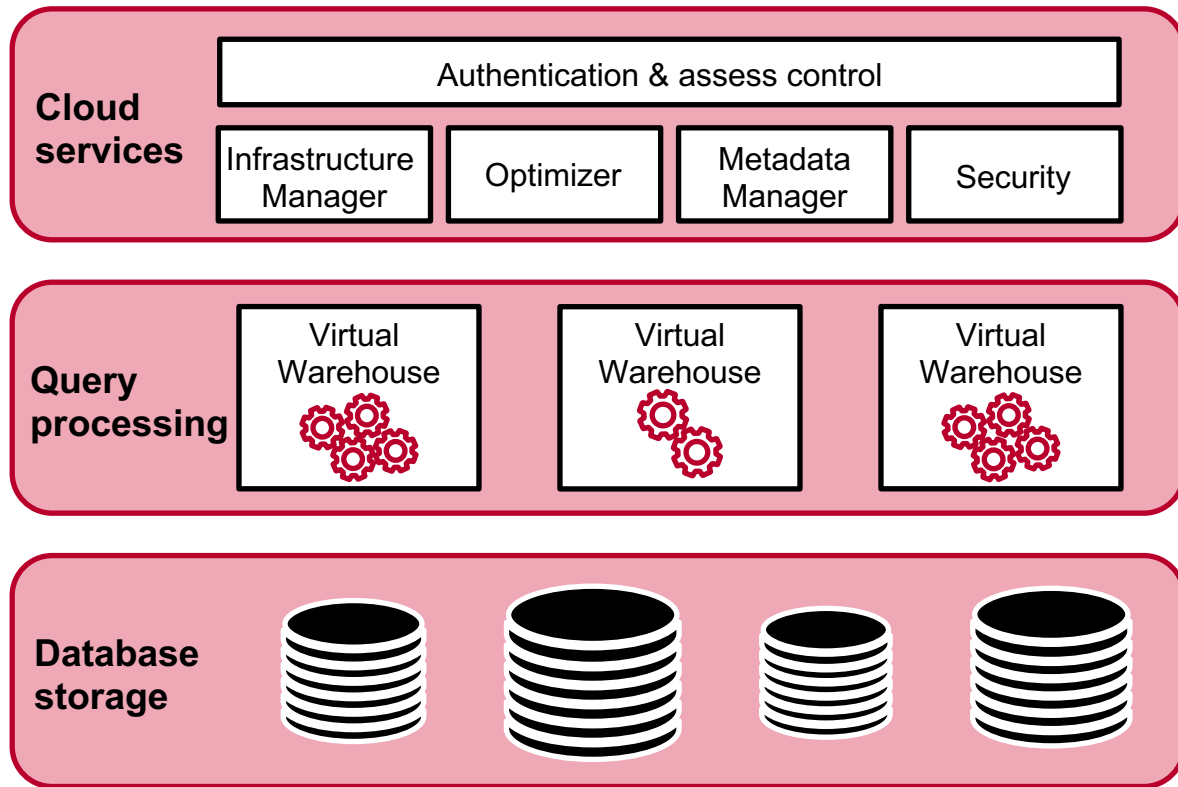
Database Storage

Query Processing

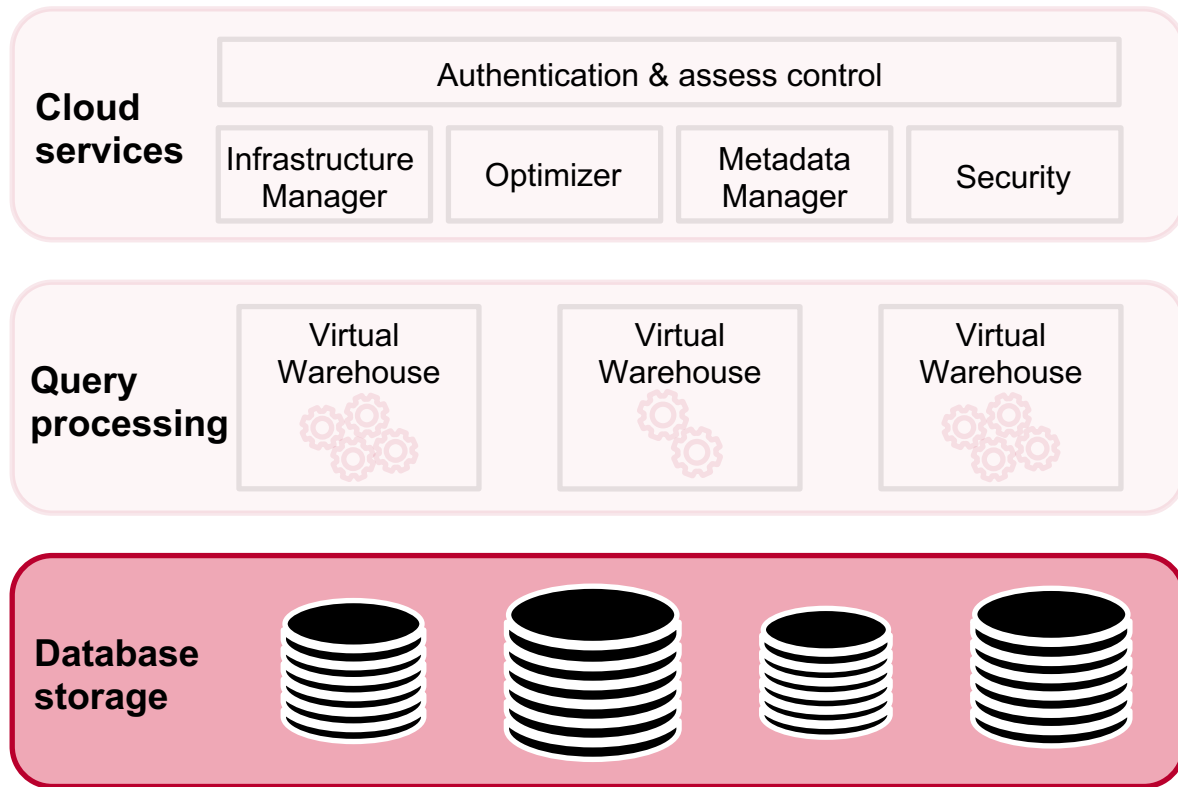
Cloud Services



Snowflake Architecture

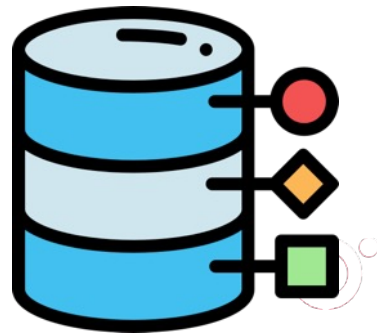


Snowflake Architecture

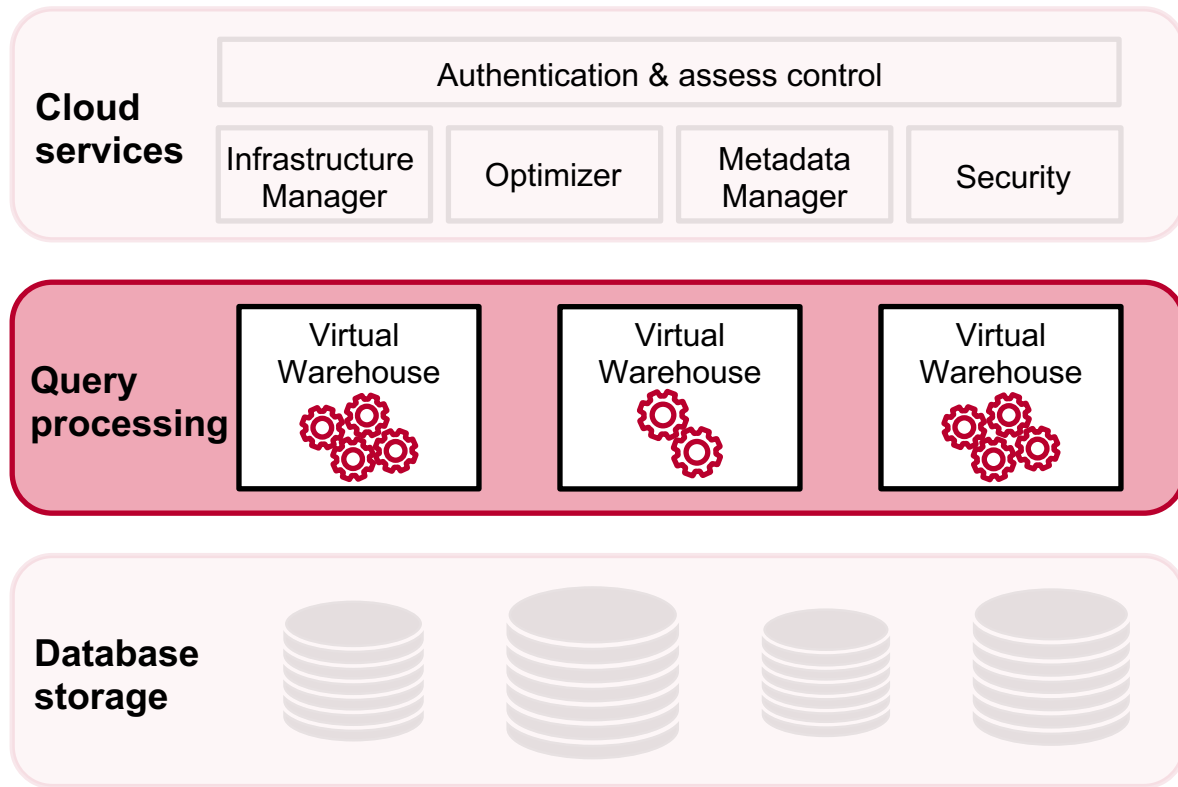


Database Storage

- Persistent storage of data loaded into Snowflake
- Proprietary internal format:
 - Columnar, compressed, optimized
- Manages all aspects of data - organization, metadata, statistics, back up
- Snowflake users cannot access or view this data directly
- Data only accessible via SQL queries

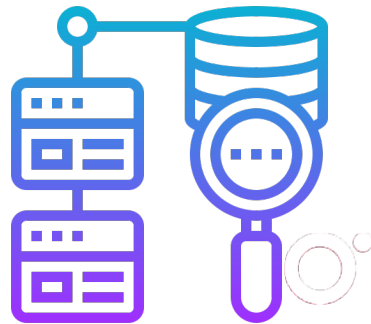


Snowflake Architecture

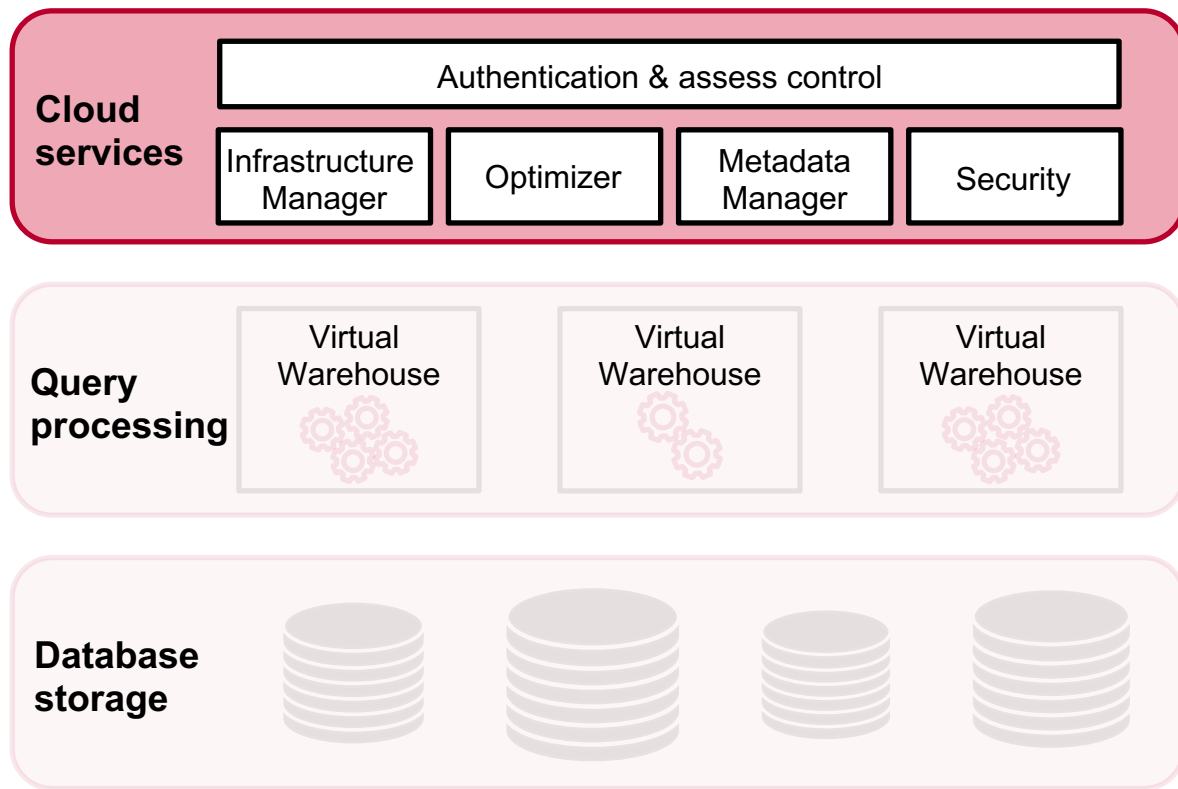


Query Processing

- Processing layer uses data warehouses
- Each warehouse a massively, parallel cluster
- Warehouses can have one or more compute nodes
- Each warehouse an **independent** compute cluster
- If queries run on different warehouses they don't compete for the same resources



Snowflake Architecture



Cloud Services

- Collection of services that coordinate activities across Snowflake
- Services run on compute instances provisioned by Snowflake on the cloud platform



Working with Snowflake

Web-based user
interface

Command-line
client (SnowSQL)

ODBC and JDBC
drivers

Native connectors
for Python, Spark
etc.

Third-party
connectors for BI,
ETL tools



Snowflake Editions

Standard

Enterprise

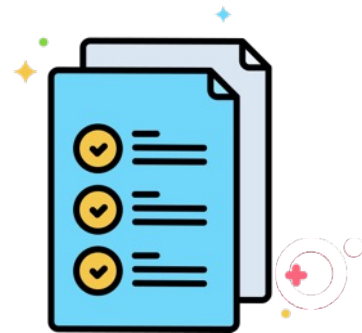
Business Critical

Virtual Private
Snowflake



Standard Edition

- Introductory offering, standard features
- Data encryption
- Object-level access control
- Time-travel up to 1 day
- Disaster recovery with fail-safe up to 7 days
- No multi-cluster warehouses, may not meet regulatory needs (government), no private connectivity



Enterprise Edition

- Includes everything in the Standard edition
- Extended time travel up to 90 days
- Additional security configurations
- May not meet regulatory needs (government), no private connectivity



Business Critical Edition

- Includes everything in the Enterprise edition
- Higher levels of data protection
- Database failover/failback for business continuity
- No dedicated metadata store and pool of compute resources



Virtual Private Snowflake

- Includes everything in the Business Critical edition
- Highest level of security for organizations with strictest requirements
- Financial institutions, government
- Snowflake environment **completely isolated** from other Snowflake accounts - **no shared resources**



Snowflake Billing Model

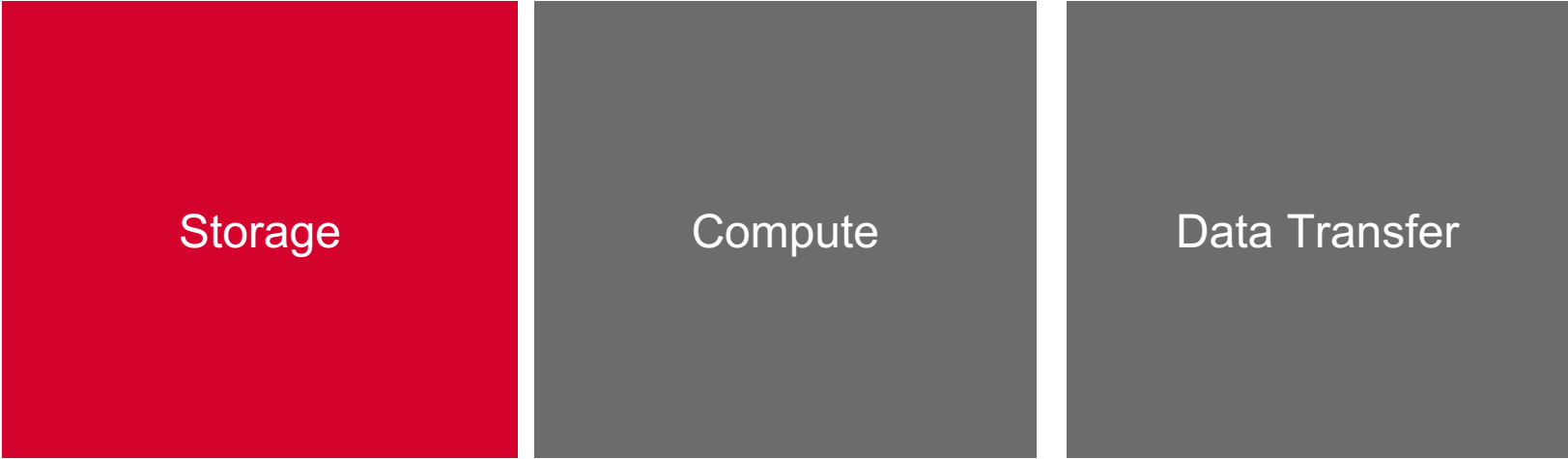
Storage

Compute

Data Transfer



Rate Per Terabyte (Data + Backups)



Storage

Compute

Data Transfer



Virtual Warehouses + Cloud Services

Storage

Compute

Data Transfer



Cloud Provider Charge for Moving Data in and Out

Storage

Compute

Data Transfer



Snowflake Credit

A unit of measure to pay for the consumption of resources on Snowflake. Credit is used only when the customer is using resources.





Poll 1

Which of the following Snowflake architecture components is responsible for query processing operations?

- Processing engine
- Virtual Warehouse
- Cloud Services
- Data Storage



Poll 1

Which of the following Snowflake architecture components is responsible for query processing operations?

- Processing engine
- Virtual Warehouse
- Cloud Services



Poll 2

Which Snowflake edition offers a dedicated pool of compute resources so your Snowflake account can be entirely isolated from other accounts?

- Standard
- Enterprise
- Business Critical
- Virtual Private Snowflake



Poll 2

Which Snowflake edition offers a dedicated pool of compute resources so your Snowflake account can be entirely isolated from other accounts?

- Standard
- Enterprise
- Business Critical

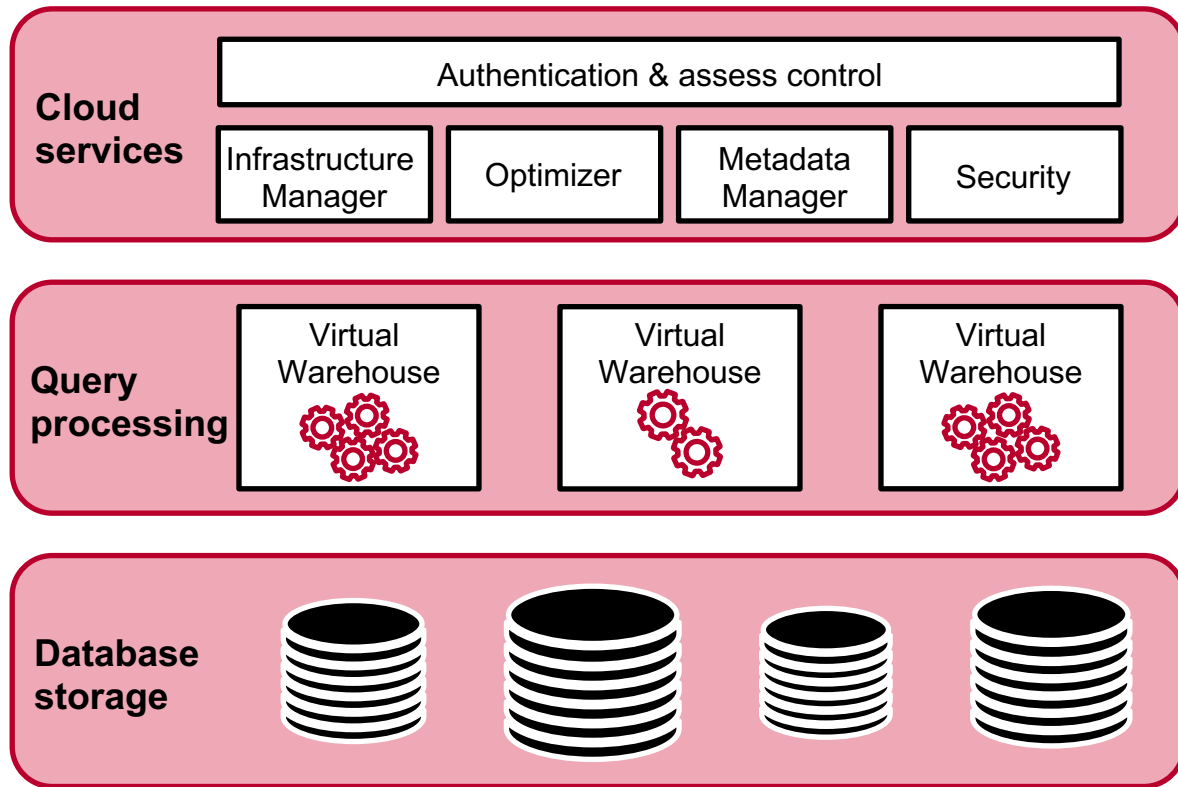


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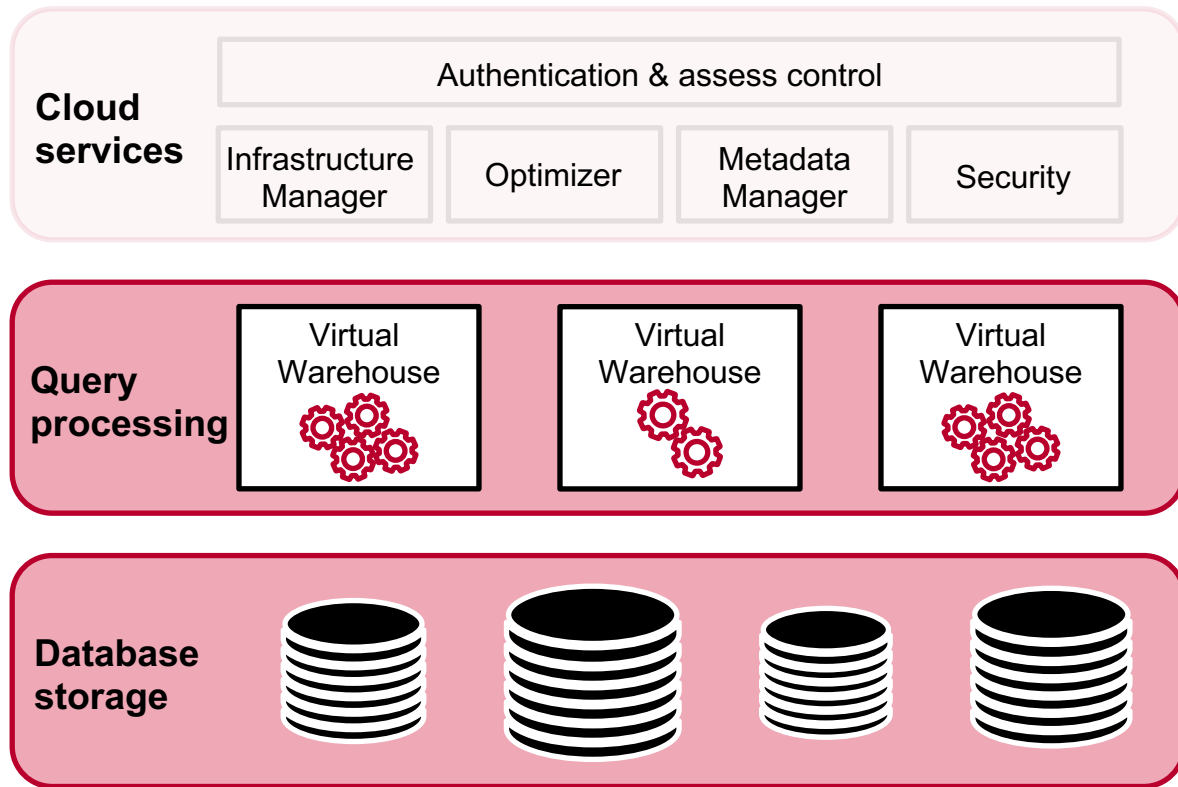
Virtual Warehouses



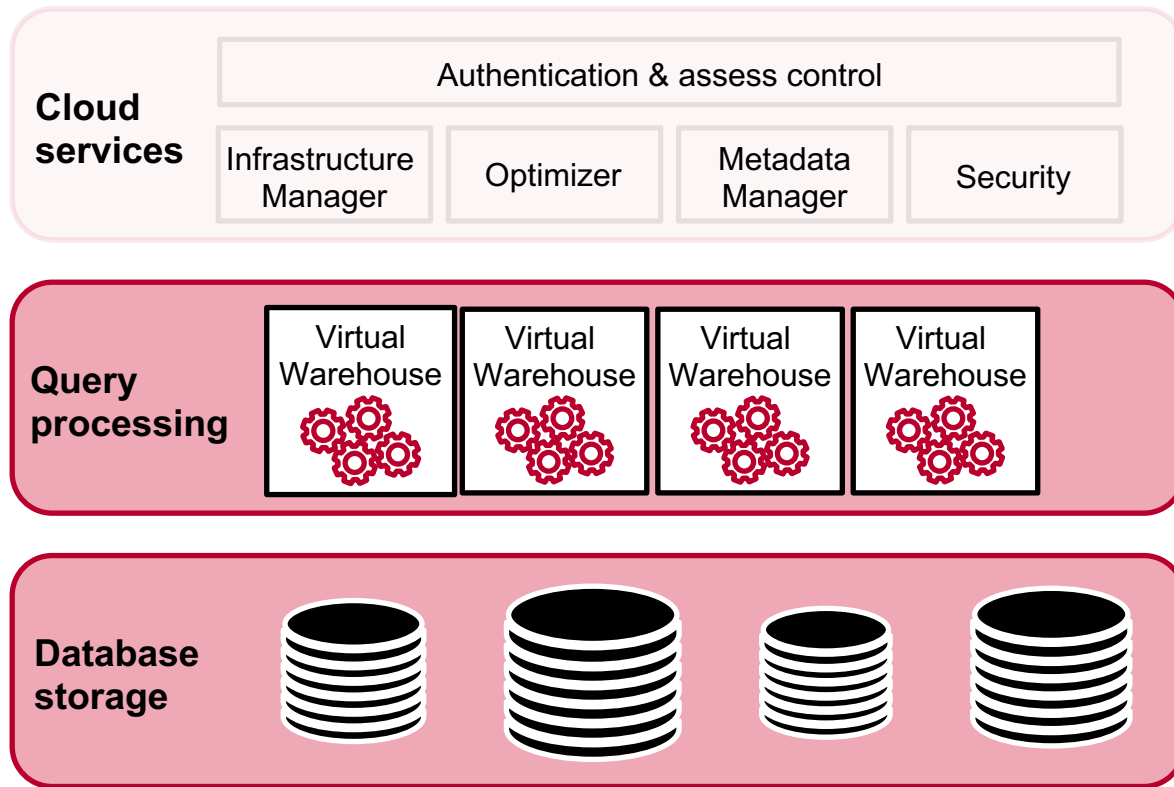
Snowflake Architecture



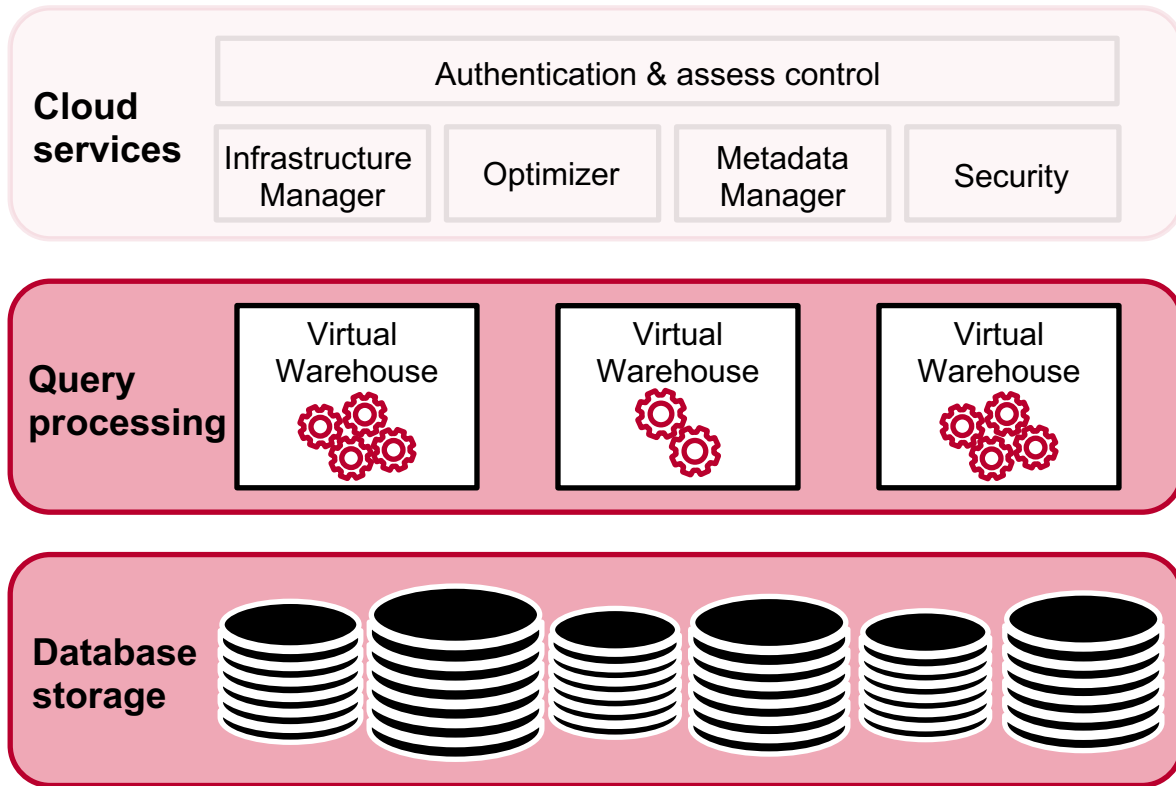
Independent Storage and Compute



Scale Compute



Scale Storage



Virtual Warehouse

Cluster of compute resources, such as CPU, memory, and temporary storage to perform operations in Snowflake



Virtual Warehouse Operations

Retrieving rows from
tables and views

Updating rows in
tables

Loading data into
tables

Unloading data from
tables



Virtual Warehouse Sizes

Warehouse Size	Credits / Hour	Credits / Second	Notes
X-Small	1	0.0003	Default size for warehouses created using CREATE WAREHOUSE .
Small	2	0.0006	
Medium	4	0.0011	
Large	8	0.0022	
X-Large	16	0.0044	Default for warehouses created in the web interface.
2X-Large	32	0.0089	
3X-Large	64	0.0178	
4X-Large	128	0.0356	
5X-Large	256	0.0711	Preview feature; currently only available on Amazon Web Services (AWS).
6X-Large	512	0.1422	Preview feature; currently only available on AWS.



Virtual Warehouse

- Size and usage of warehouse impacts **billing**
- **Data loading** performance may not depend on warehouse size
 - Also impacted by **size and number of files**
- Query processing time directly impacted by warehouse size
 - **Larger warehouses = more compute resources = faster queries** (especially complex queries)



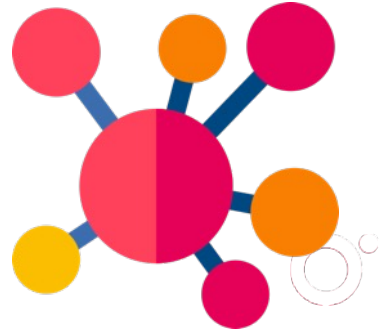
Virtual Warehouse

- Automatically suspended or resumed
- Warehouses consume credits only when they are running (not in suspended state)
- Query **concurrency** determined by size and complexity of queries
- Query is **queued** to be processed if the warehouse does not have sufficient resources



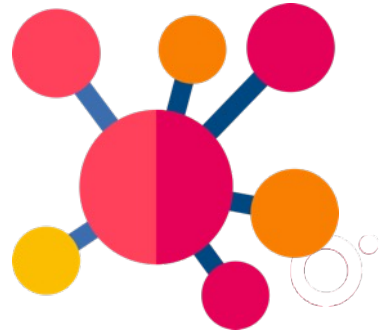
Multi-cluster Warehouses

Snowflake can statically or dynamically allocate additional warehouses to make more compute resources available



Multi-cluster Warehouses

- Snowflake automatically starts and stops additional warehouses as needed to handle **fluctuating workloads**
- Enables a large number of users to connect to the same size warehouse



Multi-cluster Warehouses

Improve
concurrency for
users and queries

Do NOT improve
the performance of
slow running
queries



Needs a Larger Warehouse

Improve
concurrency for
users and queries

Do NOT improve
the performance of
slow running
queries



Warehouse Considerations

- Charges for warehouses depend on:
 - Warehouse size
 - Number of warehouses (for multi-cluster warehouses)
 - Time the warehouse is running



Warehouse Considerations

- Compute resources for a query depends on query size and complexity
- Queries generally scale linearly w.r.t warehouse size
- Size of the tables being queried has more impact than number of rows
- Query filtering, joins, also impact processing



Best practice: Use separate
warehouses for data load
operations vs. compute
operations

Data Loading Considerations

- Load multiple files with your data - each file between 100-250MB compressed
- Loading very large files 100GB or larger is not recommended
- Loading large datasets can impact query performance
- Separate warehouses for data loading and compute
- Data loading requires **minimal resources** so warehouses can be small

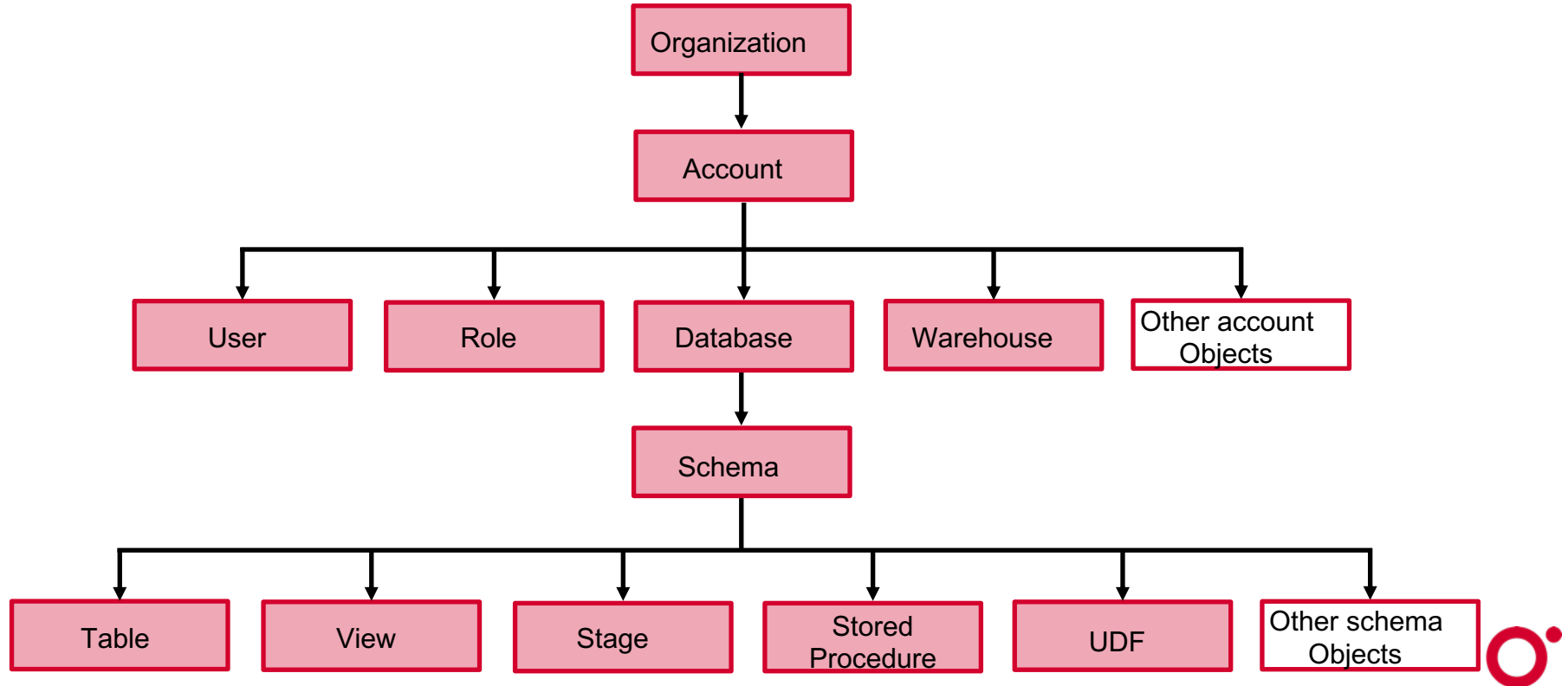


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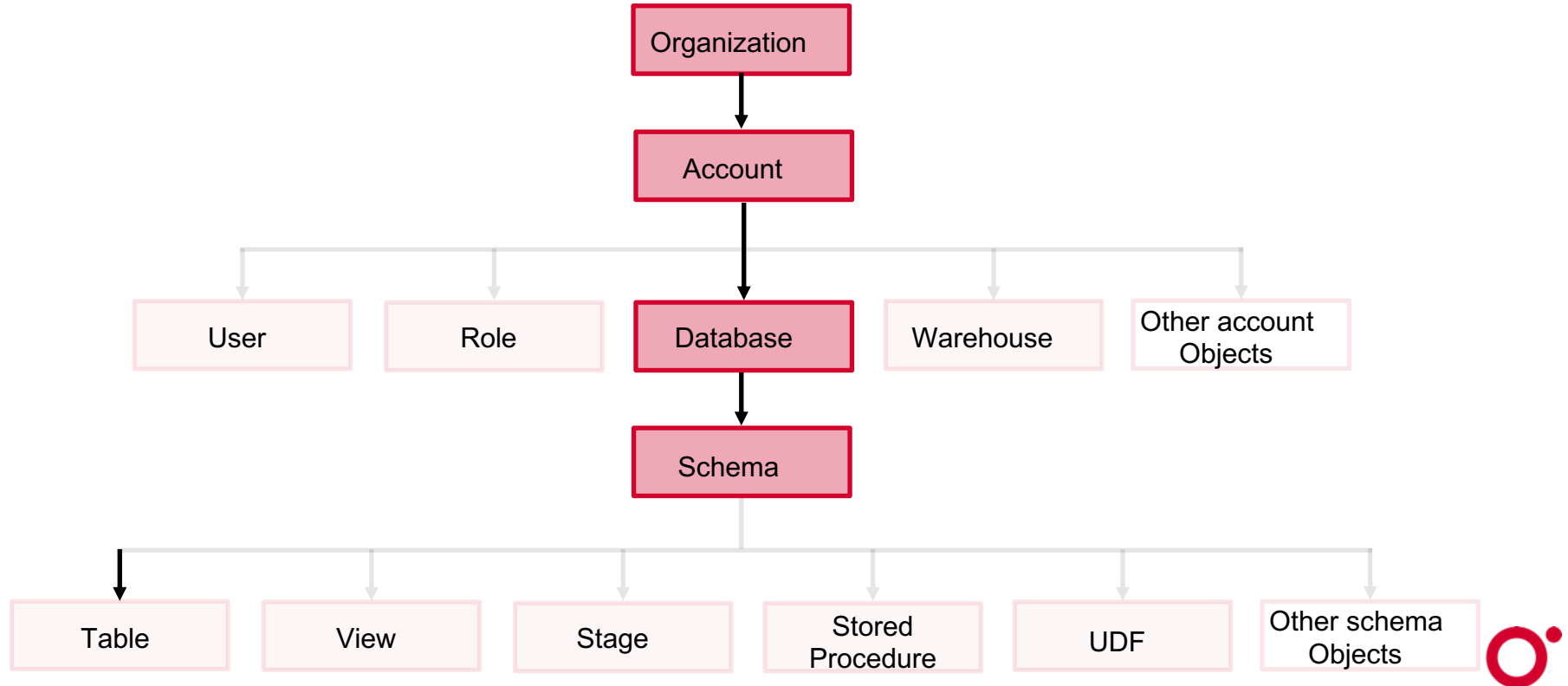
Snowflake Objects



Snowflake Hierarchy

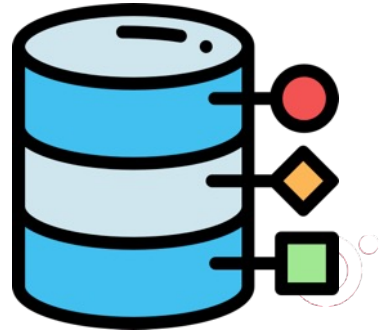


Snowflake Hierarchy



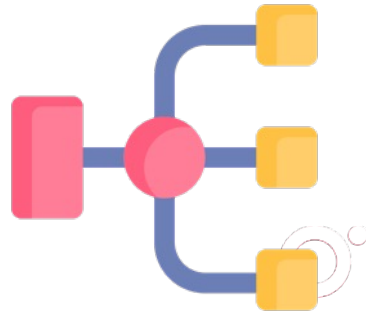
Database

Logical collection of schemas, each schema belongs to a single database

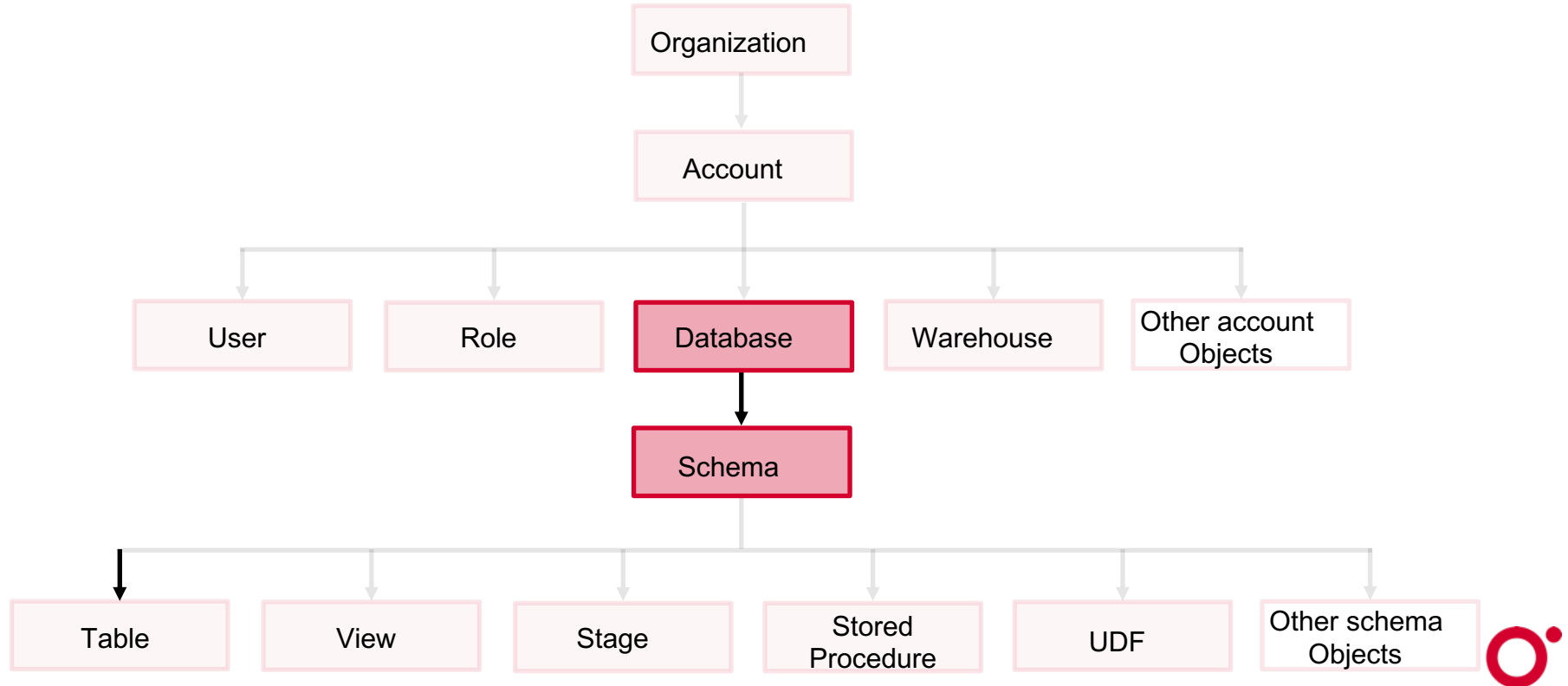


Schema

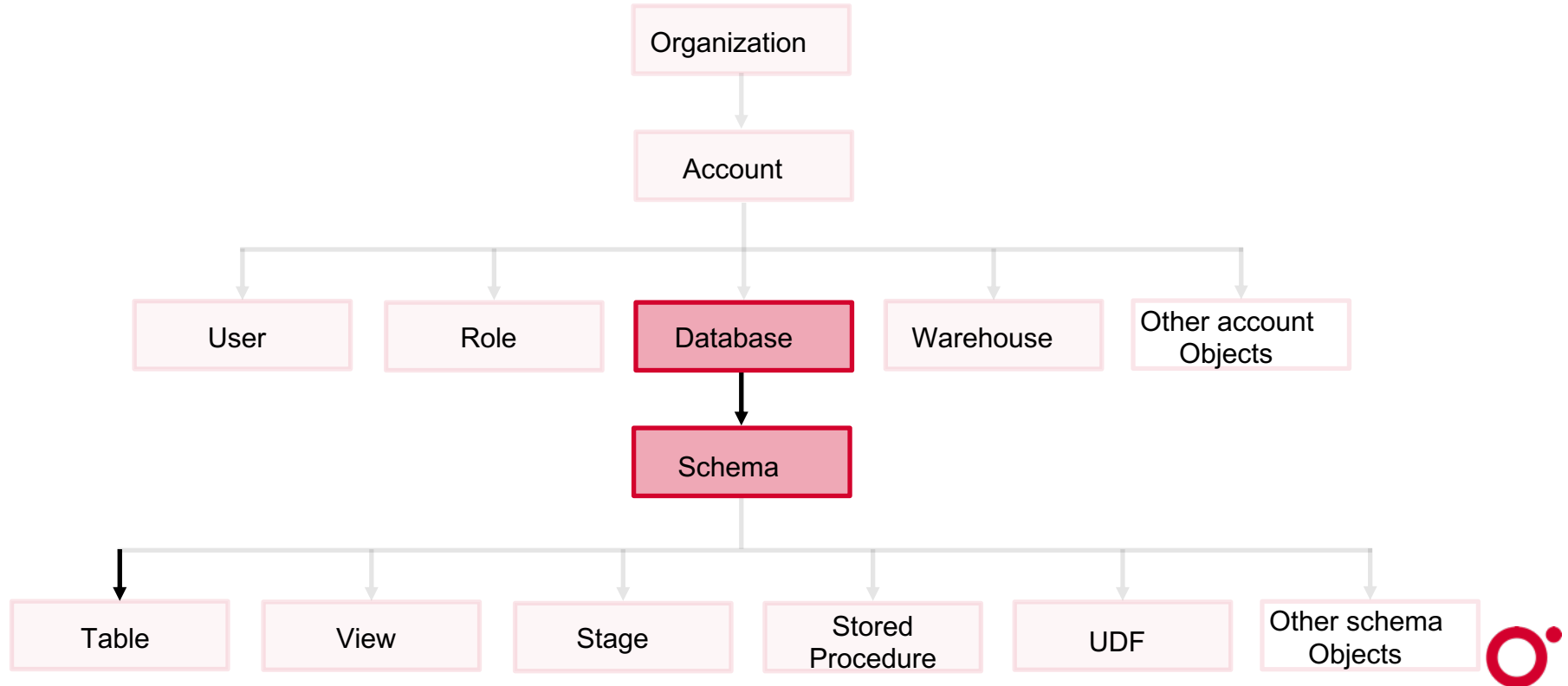
Logical grouping of database objects (tables, views, and other constructs).



Namespace = Database + Schema



Namespace Required for DB Ops





Hands-on demos:
Getting Started with Snowflake,
Databases and Warehouses,
Running Queries



Poll 3

What characteristic of Snowflake does its three layer architecture enable?

- The ability to specify database tables using a hierarchy
- The ability to extend the SQL query language for complex ops
- The ability to store data in a columnar, compressed format
- The ability to independently scale compute and storage



Poll 3

What characteristic of Snowflake does its three layer architecture enable?

- The ability to specify database tables using a hierarchy
- The ability to extend the SQL query language for complex ops
- The ability to store data in a columnar, compressed format



Poll 4

What does the schema represent in Snowflake?

- Names and data types of columns in your data
- The format of the data loaded into Snowflake
- A logical grouping of database objects
- The path that needs to be specified when accessing tables



Poll 4

What does the schema represent in Snowflake?

- Names and data types of columns in your data
- The format of the data loaded into Snowflake
- A logical grouping of database objects





Table Structures in Snowflake



Querying Big Data

Several issues when queries are run on tables that are terabytes or petabytes in size:

- Queries run slowly
- No parallelism in query processing
- Entire data needs to be scanned for each query



Better performance and
scaling is obtained using
partitioning of tables

Partitioning

A unit of management of table data. Large datasets are split into partitions which help parallelize queries and reduce the amount of data scanned.



Table Structures in Snowflake

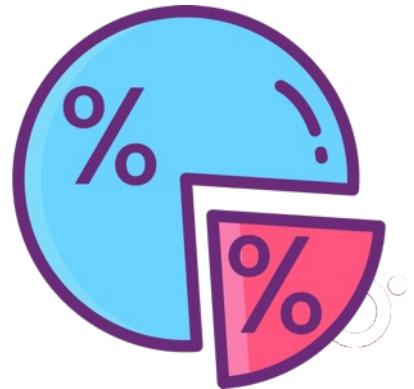
Micro-partitioning

Clustering



Micro-partitioning

- All tables divided into micro-partitions
- **Contiguous units of storage** which contain between 50MB to 500MB of uncompressed data
- Allows granular pruning of very large tables
- Snowflake stores metadata for each micro-partition
 - Range of values for each column
 - Number of distinct values
 - Additional properties for optimization and efficient query processing



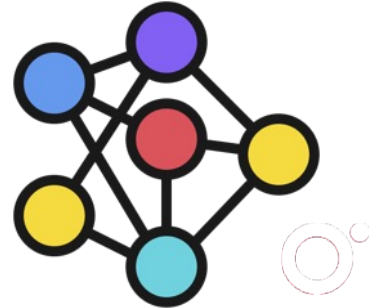
Micro-partitioning Benefits

- Automatic, no manual intervention from the user
- Fine-grained pruning == faster queries
- Uniform size, will not result in skew
- Queries run in parallel on each micro-partition
- Columnar storage (only selected columns in the result will be scanned)



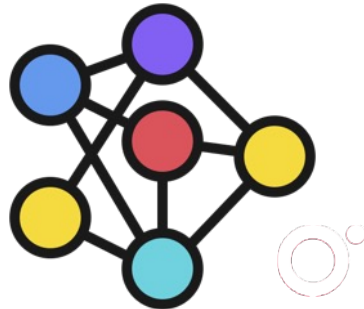
Clustering

- Data in tables **sorted along natural dimensions** based on columns i.e. clustered
- Clustering metadata recorded for each micro-partitioning when data is loaded into tables
 - Total number of micro-partitions
 - Number of micro-partitions with overlapping values
 - Depth of the overlapping micro-partitions (smaller depth = better clustering)



Clustering

- Optimize range and equality predicates of queries
- Scan only the micro-partitions which are likely to contain the required data



Data Storage in Tables

Active Data

Time Travel

Fail-safe



Data Storage in Tables

Active Data

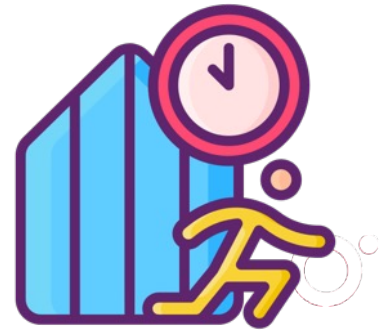
Time Travel

Fail-safe



Active Data

Data that can be actively queried. SQL queries, DDL, DML operations are all permitted on this data



Data Storage in Tables

Active Data

Time Travel

Fail-safe



Time Travel

- **Query data from the past** that has since been updated or deleted
- **Create clones** of tables, schemas, and databases at or before specific points in the past
- **Restore** tables, schemas, and databases that have been dropped



Data Storage in Tables

Active Data

Time Travel

Fail-safe



Fail-safe

- Non-configurable 7-day period during which historical data may be recoverable by Snowflake
- Period starts immediately after the time travel retention period ends



Fail-safe

- No user operations allowed
- Snowflake makes best effort to recover data
- Data recovery may take hours or days to complete



Types of Tables

Permanent

Temporary

Transient



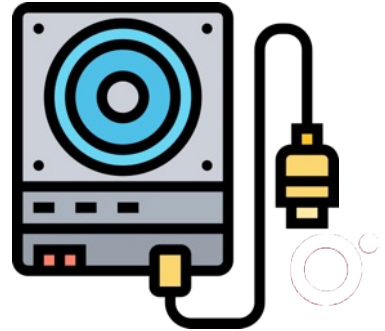
Types of Tables

Type	Persistence	Time Travel Retention Period	Fail-safe Period
Permanent (other editions)	Until explicitly dropped	0 to 90 (default configurable)	7
Permanent (Standard Edition)	Until explicitly dropped	0 or 1 (default 1)	7
Temporary	Remainder of session	0 or 1 (default 1)	0
Transient	Until explicitly dropped	0 or 1 (default 1)	0



External Tables

- Reference data files located on cloud storage
 - Amazon S3
 - Azure Data Lake
 - Google Cloud Storage
- Read-only tables, no updates allowed
- Query and join operations allowed





Hands-on demos:
Time travel
SnowSQL



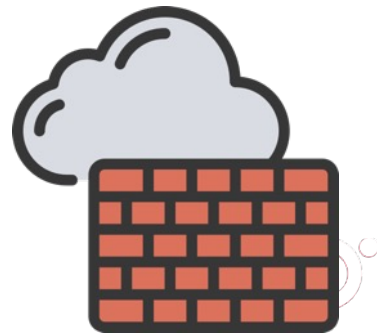


Snowflake vs. Databricks



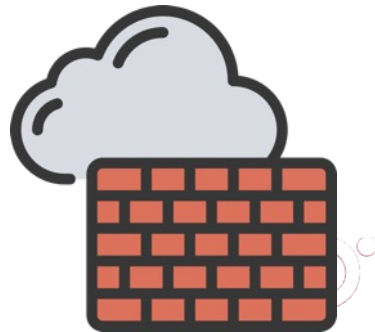
Databricks

Cloud-based data platform powered by Apache Spark for processing and analyzing big data.



Databricks

- **Databricks SQL**
 - Platform to run SQL queries to analyze data
- **Databricks Data Science and Engineering**
 - Interactive workspace for collaboration and to generate insights using Spark
- **Databricks Machine Learning**
 - Integrated end-to-end machine learning environment with managed services



Databricks Key Features

Delta Lake for the entire
data lifecycle

Optimized Spark Engine
for data processing

Machine Learning
Environments

Collaborative Notebooks
for Scala R, SQL, Python



Snowflake vs. Databricks

Snowflake

- Managed data platform
- Integrated with the 3 major cloud platforms
- Decoupled storage and processing layers
- Highly scalable

Databricks

- Managed data platform
- Integrated with the 3 major cloud platforms
- Decoupled storage and processing layers
- Highly scalable



Snowflake vs. Databricks

Snowflake

- SQL-based so **easier and intuitive** to use
- Data processing with Spark using a connector
- Less versatile in terms of use cases (no ML)
- Batch-centric but can work with continuous data

Databricks

- SQL + R, Python, Scala more options to process data
- Spark at the core of the Databricks ecosystem
- **More versatile** (SQL queries, data processing, ML)
- Strong **batch and streaming** support (with Spark)



Poll 5

Which of the following is NOT a data storage component in Snowflake?

- Backup
- Fail-safe
- Time travel
- Active or current data



Poll 5

Which of the following is NOT a data storage component in Snowflake?

- Backup
- Fail-safe
- Time travel



Poll 6

Which of the following types of tables is only available within a single session?

- Permanent
- Temporary
- Transient



Poll 6

Which of the following types of tables is only available within a single session?

- Permanent
- Temporary
- Transient



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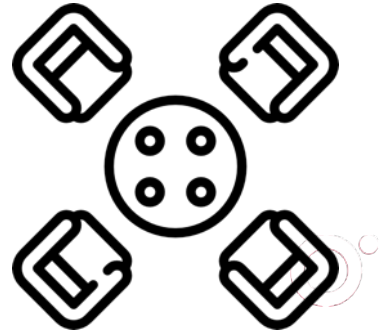
Regular, Secure, Materialized
Views



View

A query which produces a result, this result can then be accessed as a table.

A view helps combine, segregate, or protect data



Views in Snowflake

Non-materialized
views

Materialized
views



Non-materialized Views

Named definition of a query.

When a view is referenced in another query the view's query is executed. The view's results are not stored for future use.

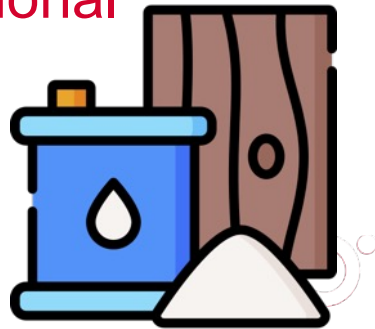


Materialized Views

Named definition of a query, however results of the view's query are stored as though they were a table.

When a view is referenced in another query that uses the stored query results of the materialized view.

Queries run faster but the stored results take up additional space



Secure Views



Secure Views

Views with improved data privacy and sharing

Does not allow query optimizations that might inadvertently expose data from the underlying table to the user.





Hands-on demos:
Views



Poll 7

Which of the following is true for a user-defined function?

- Has to be invoked as a part of a SQL query
- May or may not return a value
- Allows recursive calls to itself



Poll 7

Which of the following is true for a user-defined function?

- Has to be invoked as a part of a SQL query
- May or may not return a value
- Allows recursive calls to itself



Poll 8

Which of the following statements about views is true?

- Queries on a secure view run faster than queries on regular views
- Queries on a materialized view run faster than queries on regular views
- Queries on regular views run faster than queries on materialized views



Poll 8

Which of the following statements about views is true?

- Queries on a secure view run faster than queries on regular views
- Queries on a materialized view run faster than queries on regular views
- Queries on regular views run faster than queries on materialized views



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Sampling

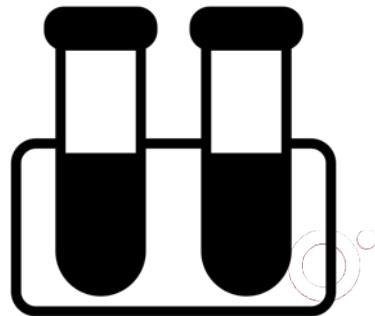


Sampling

Return a subset of rows from the specified table

Snowflake allows you to:

- Sample a fraction of the table with a specified probability “p” for including a certain row
- Sample a fixed, specified number of rows



Sampling in Snowflake

Bernoulli or Row
sampling

System or Block
sampling

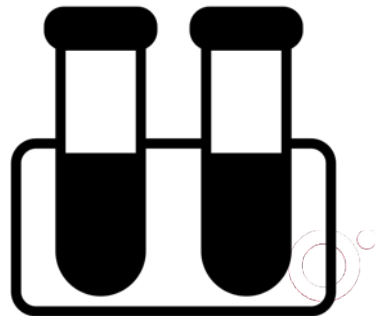


Bernoulli Sampling

Samples each row with a certain probability = $p/100$

Works well for smaller datasets as it reduces the bias of sampled data.

Is slower for larger datasets because each row considered and included with a certain probability



Block Sampling

Samples each block with a certain probability = $p/100$

May result in biased samples for smaller datasets

Works well for larger dataset, is faster and more efficient





Hands-on demos:
Sampling
Common Table Expressions



Poll 9

If you have a very large table what kind of sampling technique would you choose for that table for fast and efficient sampling?

- Bernoulli sampling
- Random sampling
- Block sampling



Poll 9

If you have a very large table what kind of sampling technique would you choose for that table for fast and efficient sampling?

- Bernoulli sampling
- Random sampling
- Block sampling



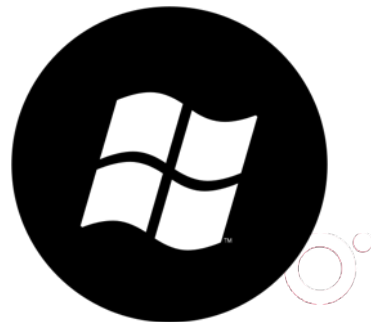
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Window Functions



Window Functions

Operate on windows - which are groups of rows that are related to one another.



Window Functions

Department	Name	Salary
Engineering	Carrie	7000
Sales	Dora	5000
Engineering	Jackson	6600
Sales	Stewart	6200
Sales	Ruby	4000
Engineering	Sonia	6850
Sales	Ivan	4200
Engineering	Steven	6550



Highest Salary in Department

Department	Name	Salary
Engineering	Carrie	7000
Sales	Dora	5000
Engineering	Jackson	6600
Sales	Stewart	6200
Sales	Ruby	4000
Engineering	Sonia	6850
Sales	Ivan	4200
Engineering	Steven	6550



Partition by Department

Department	Name	Salary
Engineering	Carrie	7000
Sales	Dora	5000
Engineering	Jackson	6600
Sales	Stewart	6200
Sales	Ruby	4000
Engineering	Sonia	6850
Sales	Ivan	4200
Engineering	Steven	6550



Partition by Department

Department	Name	Salary
Engineering	Carrie	7000
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Engineering	Steven	6550
Sales	Dora	5000
Sales	Stewart	6200
Sales	Ruby	4000
Sales	Ivan	4200



Order by Salary DESC

Department	Name	Salary
Engineering	Carrie	7000
Engineering	Jackson	6600
Engineering	Sonia	6850
Engineering	Steven	6550
Sales	Dora	5000
Sales	Stewart	6200
Sales	Ruby	4000
Sales	Ivan	4200



Order by Salary DESC

Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000



Highest Salary in Department

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Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
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Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000



Highest Salary in Department

Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000



Partition By Department Order by Salary

Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000



Partition By Department Order by Salary



Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000

ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW



Partition By Department Order by Salary



Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000

ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW



Partition By Department Order by Salary

Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000



ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW



Partition By Department Order by Salary

Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000



ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW



Partition By Department Order by Salary



Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000

ROWS BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING



Partition By Department Order by Salary



Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000

ROWS BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING



Partition By Department Order by Salary

Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
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ROWS BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING



Partition By Department Order by Salary

Department	Name	Salary
Engineering	Carrie	7000
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ROWS BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING



Partition By Department Order by Salary



Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000

ROWS BETWEEN 1 PRECEDING AND 1 FOLLOWING



Partition By Department Order by Salary



Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000

ROWS BETWEEN 1 PRECEDING AND 1 FOLLOWING



Partition By Department Order by Salary

Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000



ROWS BETWEEN 1 PRECEDING AND 1 FOLLOWING



Partition By Department Order by Salary

Department	Name	Salary
Engineering	Carrie	7000
Engineering	Sonia	6850
Engineering	Jackson	6600
Engineering	Steven	6550
Sales	Stewart	6200
Sales	Dora	5000
Sales	Ivan	4200
Sales	Ruby	4000



ROWS BETWEEN 1 PRECEDING AND 1 FOLLOWING





Hands-on demos:
Window functions



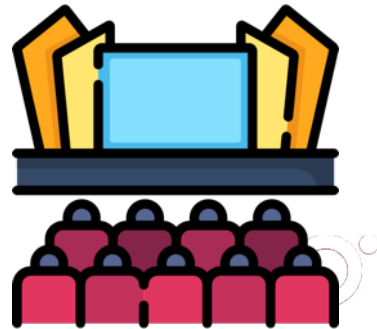
O'REILLY[®]

Data Loading in Snowflake



Stage

Refers to the location of data files in cloud storage as a stage.



Types of Stages

External Stage

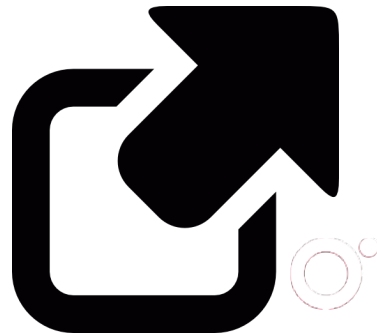
Internal Stage



External Stage

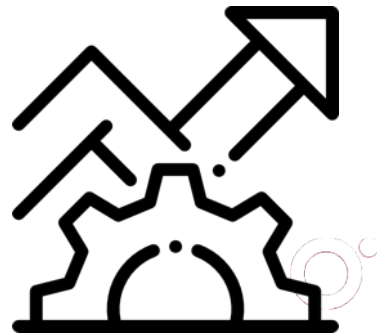
Cloud storage services that belong to the cloud platform that you are using:

- Amazon S3
- Azure Storage
- Google Cloud Storage



Internal Stages

Cloud storage that belongs to your Snowflake account



Internal Stages

User Stage

Table Stage

Named Stage



User Stage

- Stage associated with a user
- Data in stage can be loaded into multiple tables

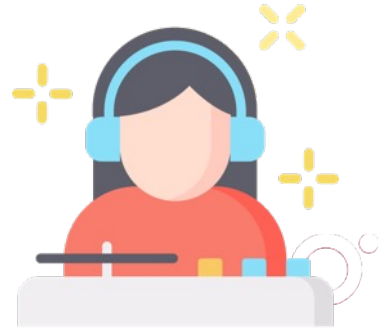
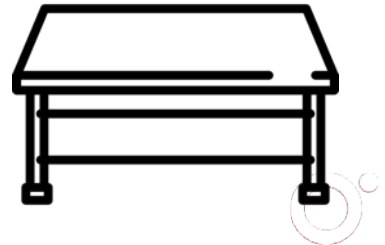


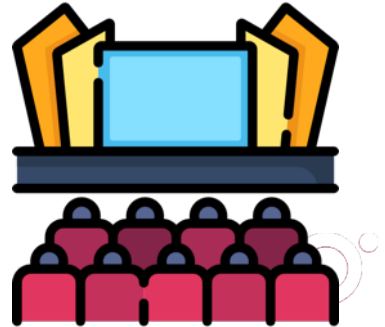
Table Stage

- Implicit stage associated with a table
- Stage can be managed by one or more users
- Data in this stage can only be loaded into the corresponding table



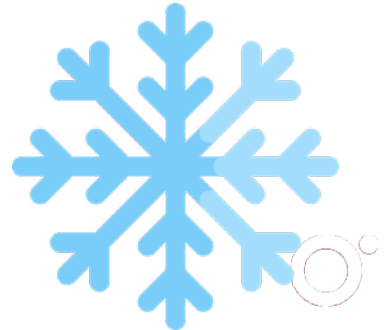
Named Stage

- A database object created in a schema
- Used and managed by one or more users
- Data in stage can be loaded into one or more tables



Snowpipe

- Snowflake's continuous data ingestion service
- Loads data added to stages into Snowflake (in minutes)
- Provides a pipelines for loading fresh data in micro-batches as soon as data is available





Hands-on demos:
Bulk Loading
Continuous Loading



Poll 10

If you wanted the data loaded in your stage to be managed by multiple users and loaded into any number of tables, what stage would you use?

- Table stage
- User stage
- Named stage



Poll 10

If you wanted the data loaded in your stage to be managed by multiple users and loaded into any number of tables, what stage would you use?

- Table stage
- User stage
- Named stage





Hands-on demos:
Semi-structured Data

